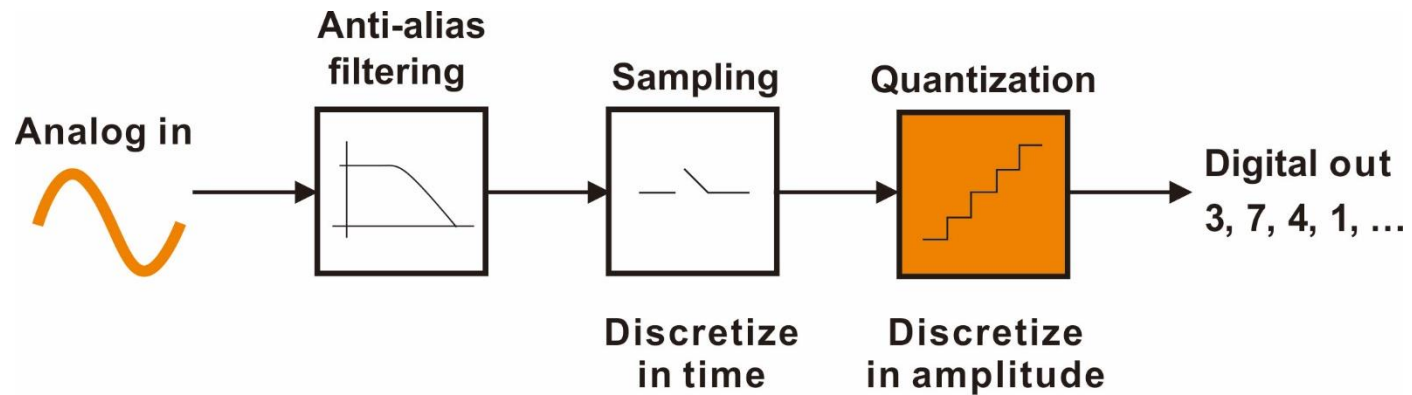


Voltage Comparators

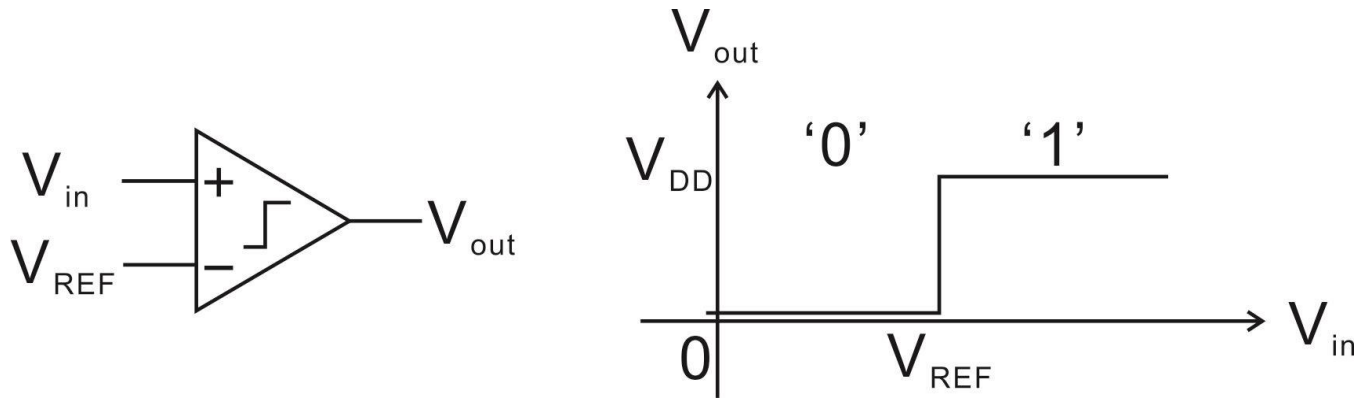
Nan Sun
Tsinghua University
nansun@Tsinghua.edu.cn

ADC Big Picture



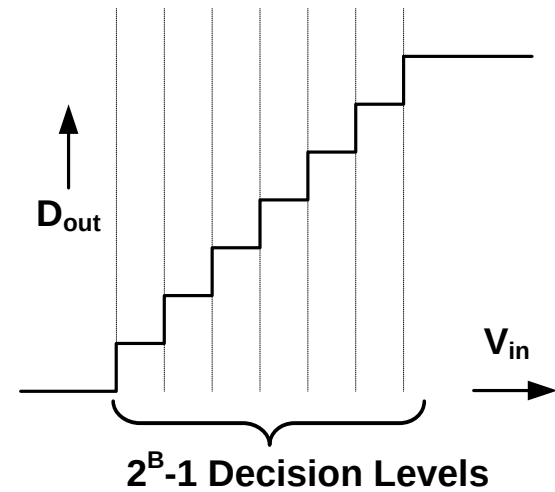
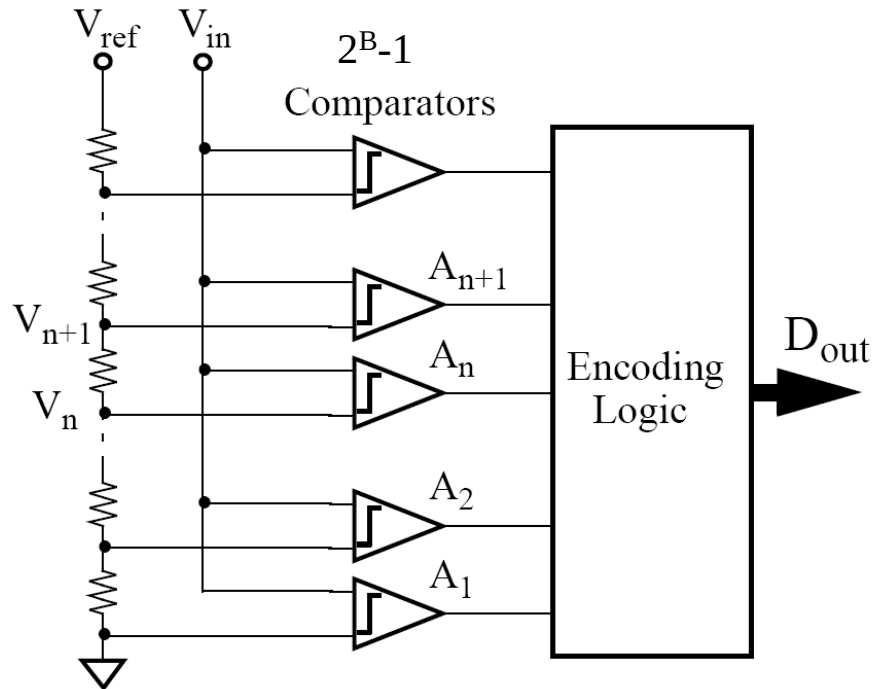
- How to quantize a voltage signal?
 - Use voltage comparators to make decisions

Ideal Voltage Comparator



- The comparator acts as a 1-bit ADC. It converts an “analog” input into a “digital” output.
- How to get more bits?
 - Use an array of comparators

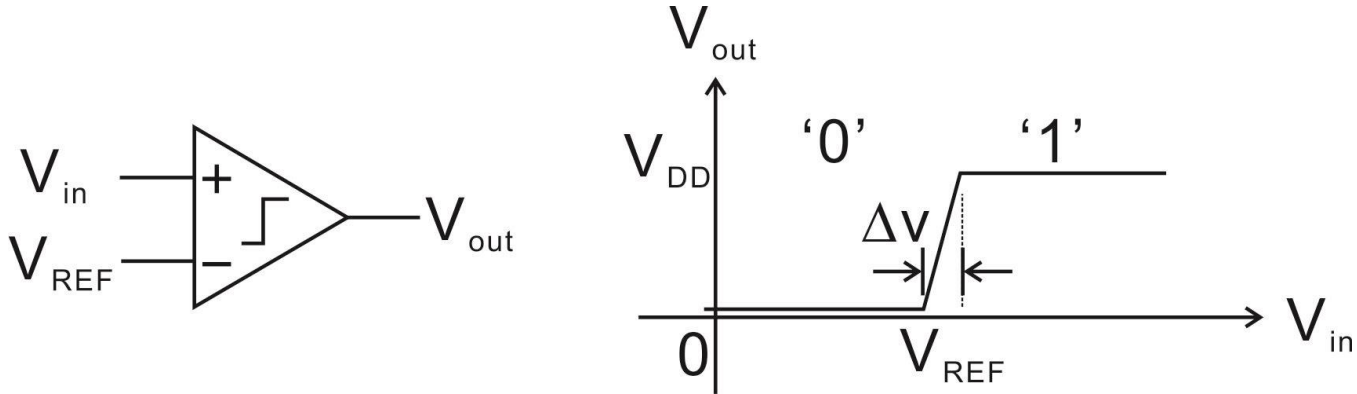
Flash ADC Basic Concept



Design Considerations

- Accuracy
 - Comparator resolution
 - Comparator offset
- Speed
 - Comparator delay
 - Comparator recovery time
- Power dissipation
- Other properties
 - Common-mode rejection
 - Input capacitance and linearity of input capacitance
 - Kickback noise

Gain Requirements



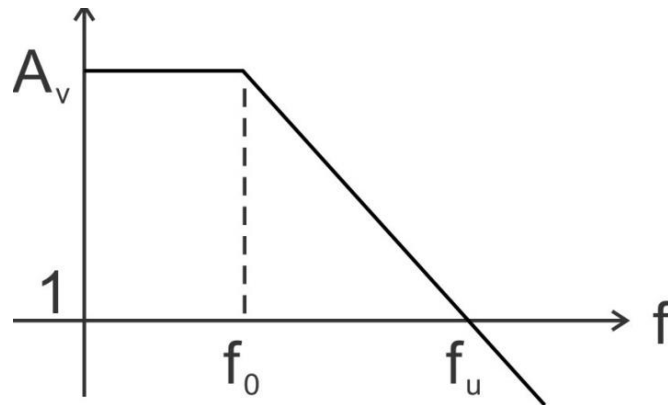
- Comparator output is 'digitally' unknown for $V_{REF} - \Delta/2 < V_{in} < V_{REF} + \Delta/2$
- We need comparator with a large effective gain A_v
- Assuming $\Delta V = 1\text{mV}$ and $V_{DD} = 1.8\text{V}$, we have:

$$A_v = \frac{1.8\text{V}}{1\text{mV}} = 1800$$

- How can we achieve this high gain?

How about Using an Op-Amp?

- Opamp can achieve high gain, but its speed is a problem.
- For an opamp, $A_v = g_m R$, $\omega_0 = 1/RC \rightarrow A_v \omega_0 = g_m/C = I_d/C * (g_m/I_d)$
 - g_m/I_d depends on biasing, but usually around $10 \sim 20 \text{ V}^{-1}$
 - For the same I_d (power), a direct trade-off between A_v and f_0



E.g., $f_u = A_v f_0 = 1\text{GHz}$ and $A_v = 1800$

$$\Rightarrow f_0 = \frac{f_u}{A_v} \approx \frac{1\text{GHz}}{1800} \approx 500 \text{ kHz}$$

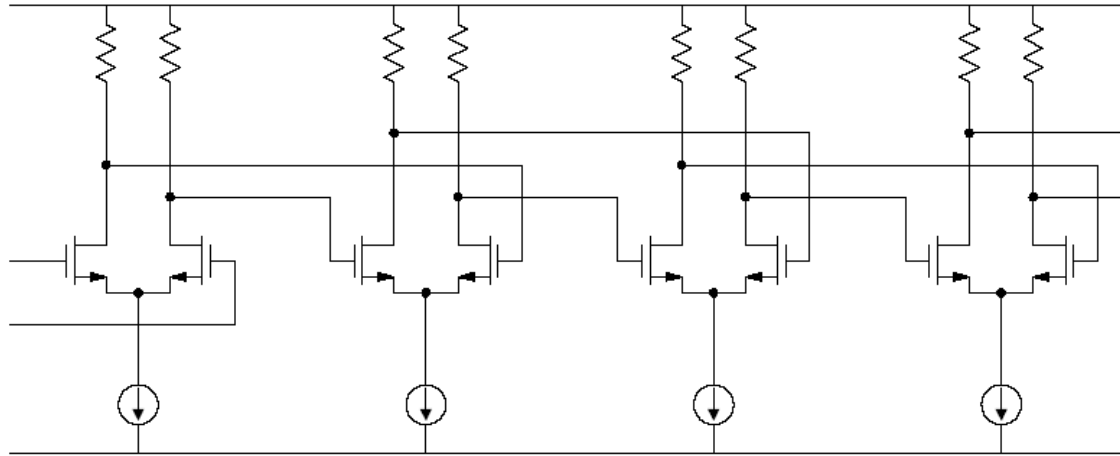
Too slow!

- We can trade power for speed: 2X power $\rightarrow$ 2X speed
- Is there a smarter approach?

Cascade 2 Amplifiers

- Analysis
 - 2 amplifier $\rightarrow$ 2X power
 - Total gain $A_v \rightarrow$ each amplifier with gain of $A_v^{0.5}$ and BW of $A_v^{0.5}f_0$
 - Thus, the overall bandwidth $\approx A_v^{0.5}f_0 \gg 2f_0$
 - Much favorable tradeoff!
- Can cascade even more

Cascade of Amplifiers



In each stage: $A_0 = A_v^{1/N} \quad \omega_0 = \frac{\omega_u}{A_0}$

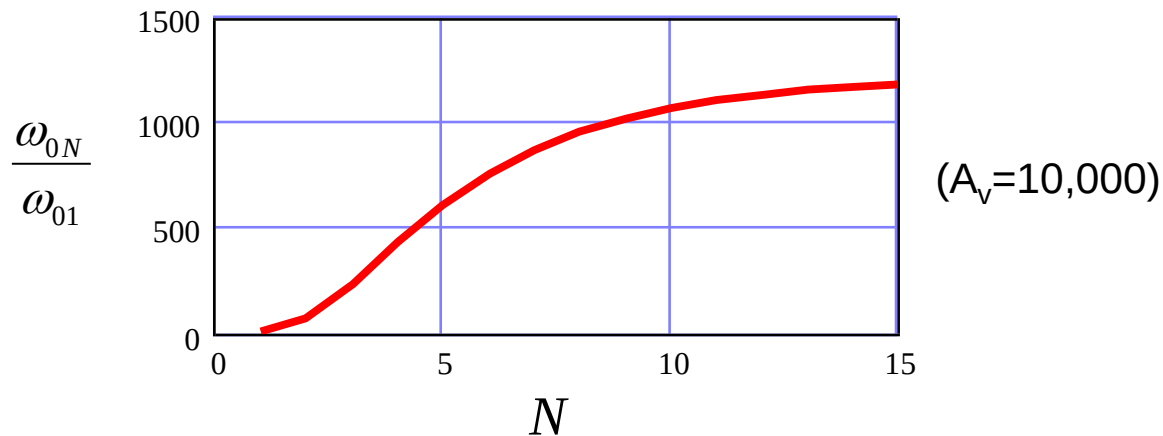
Overall: $A_v(s) = \frac{A_v}{(1 + s / \omega_0)^N} \quad \omega_{0N} = \frac{\omega_u}{A_0} \sqrt{2^{1/N} - 1}$

A_v : Total gain requirement

ω_{0N} : 3dB bandwidth of N stage realization

Bandwidth Perspective

$$\omega_{0N} = \omega_u \frac{\sqrt{2^{1/N} - 1}}{A_V^{1/N}}$$

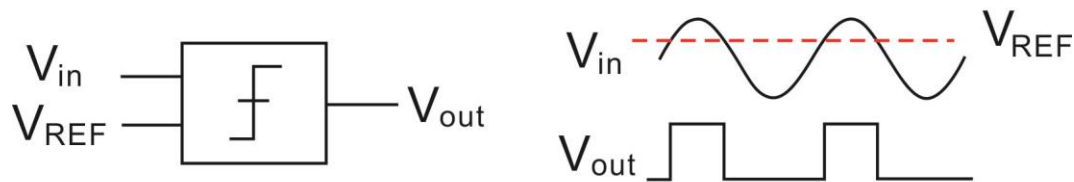


- As N increases, the return in BW diminishes and the power goes up. We do not want too large N .
- We also do not prefer a large N from the delay perspective.

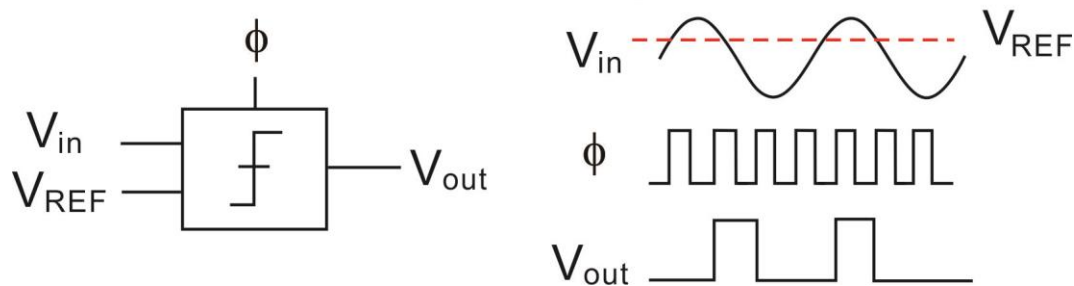
Bandwidth vs. Delay

- What we care most is the comparator delay.
- Comparators in ADCs are usually clocked.
 - Comparison starts when $\phi = 1$; reset when $\phi = 0$
- Analysis: the input is equivalent to a step

Continuous-time comparison



Clocked comparison



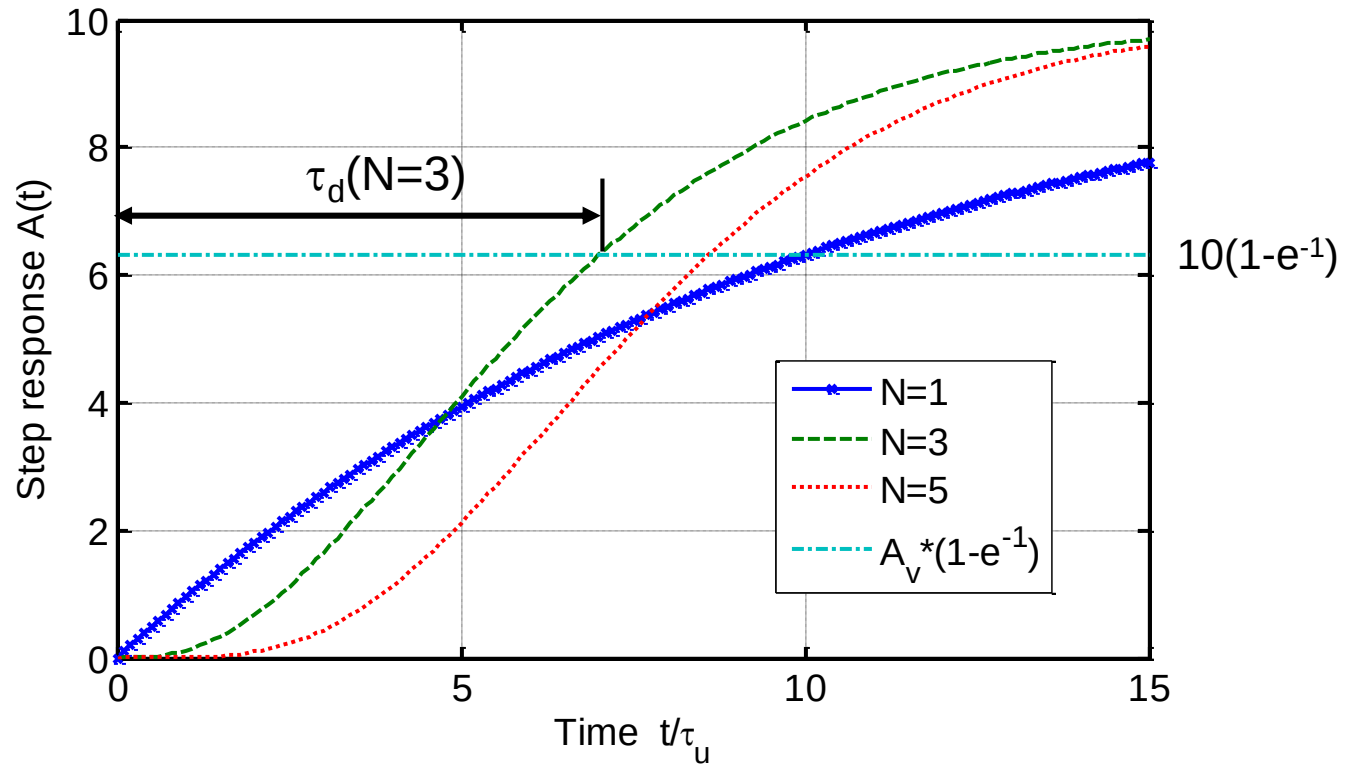
Step Response (1)

- Calculate the comparator delay for a step input:

$$V_{out}(s) = V_{in}(s)A(s) = \frac{V_{step}}{s} \frac{A_V}{(1 + s \cdot \tau_u A_V^{1/N})^N} \quad \tau_u = \frac{1}{\omega_u}$$

$$V_{out}(t) = V_{step} A_V \underbrace{\left(1 - e^{\frac{-t}{\tau_u A_V^{1/N}}} \sum_{k=0}^{N-1} \frac{\left(\frac{t}{\tau_u A_V^{1/N}} \right)^k}{k!} \right)}_{A(t)}$$

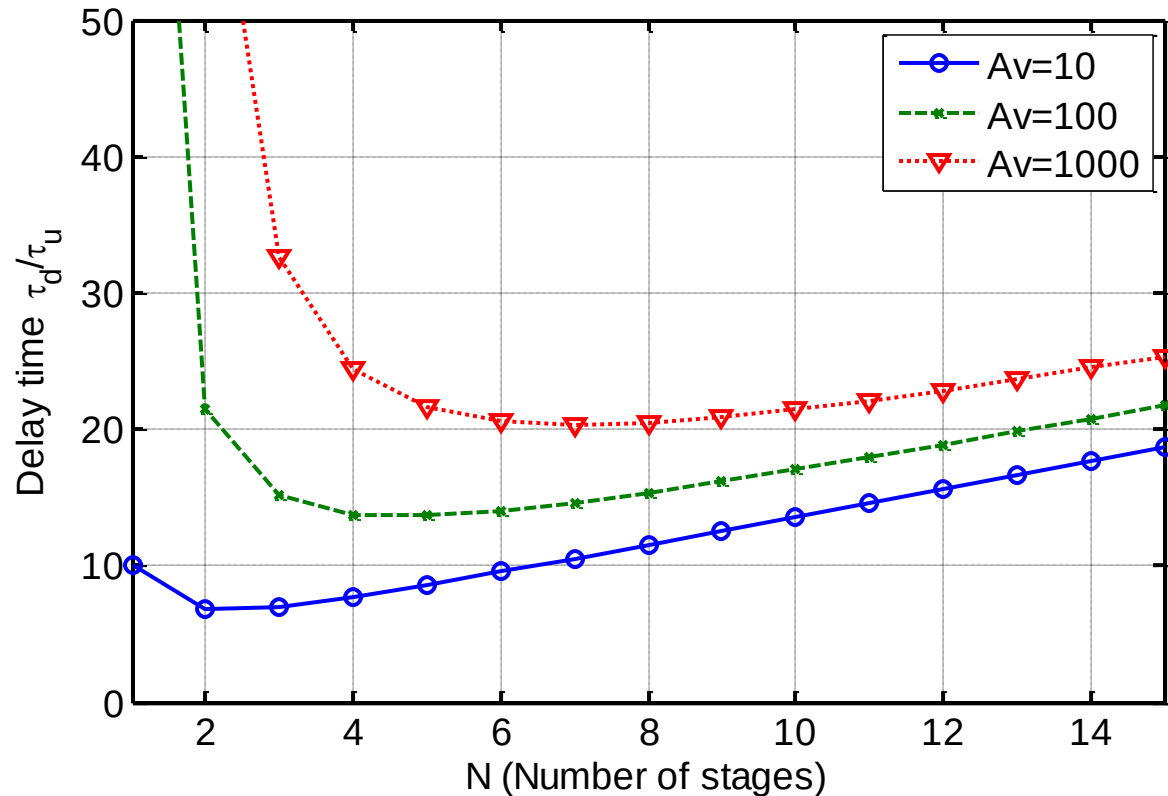
Step Response (2)



- 3-stage amplifier wins for $A_V=10$!

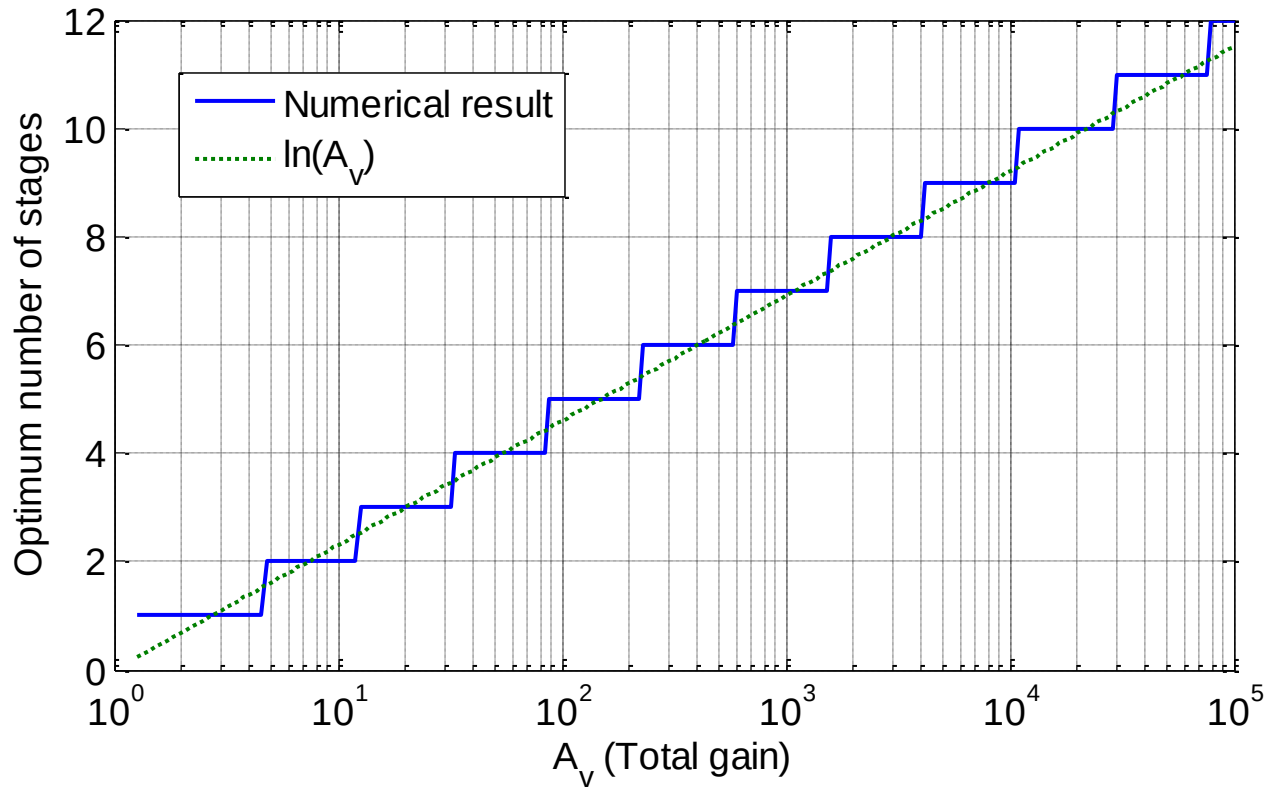
Delay vs. Number of Stages

$$A(\tau_d) = A_v(1 - e^{-1}) \Rightarrow \tau_d$$



- The minima are shallow

Optimum Number of Stages



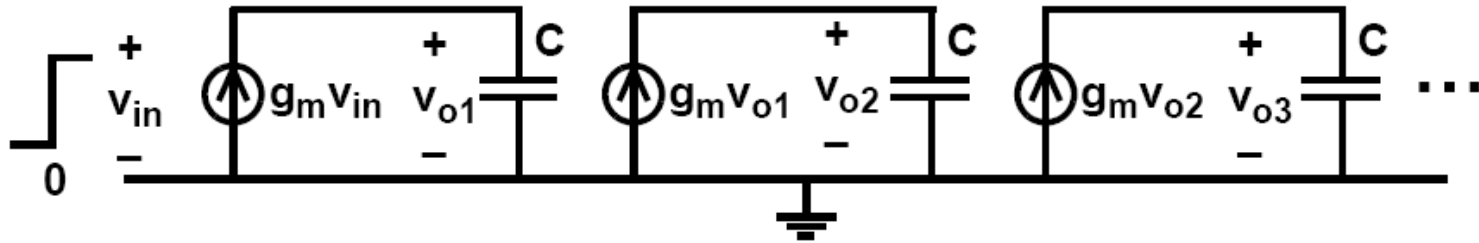
$$N_{opt} \approx \ln(A_V)$$

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Can We Further Increase Speed?

- Cascade multi-stage integrators
- Regenerative latch using positive feedback
 - E.g. cross coupled inverters

Cascade of Integrators



- Intuition
 - Load resistors (in cascade of open loop amplifiers) shunt signal current away from load capacitance → this slows down the amplification
 - Let us use all signal currents to charge the load capacitor
- Step response analysis

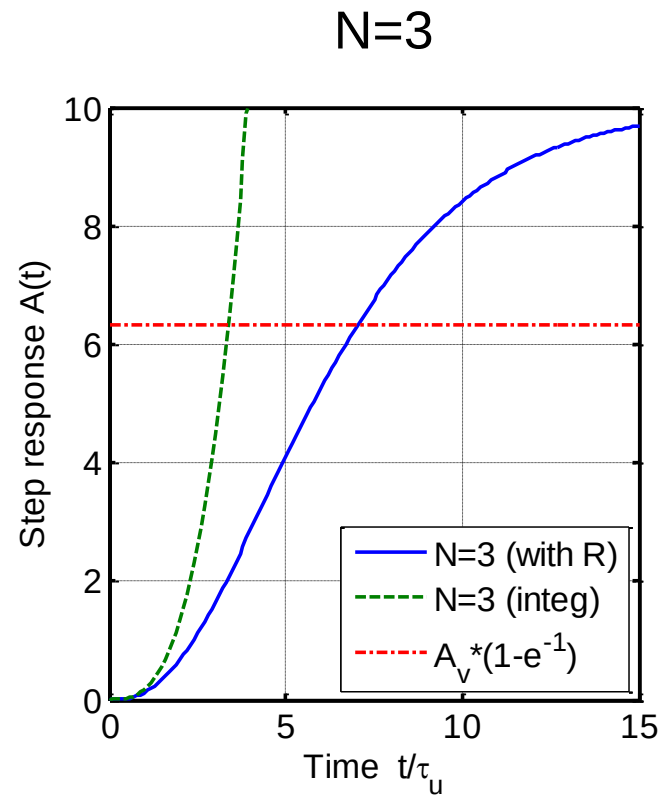
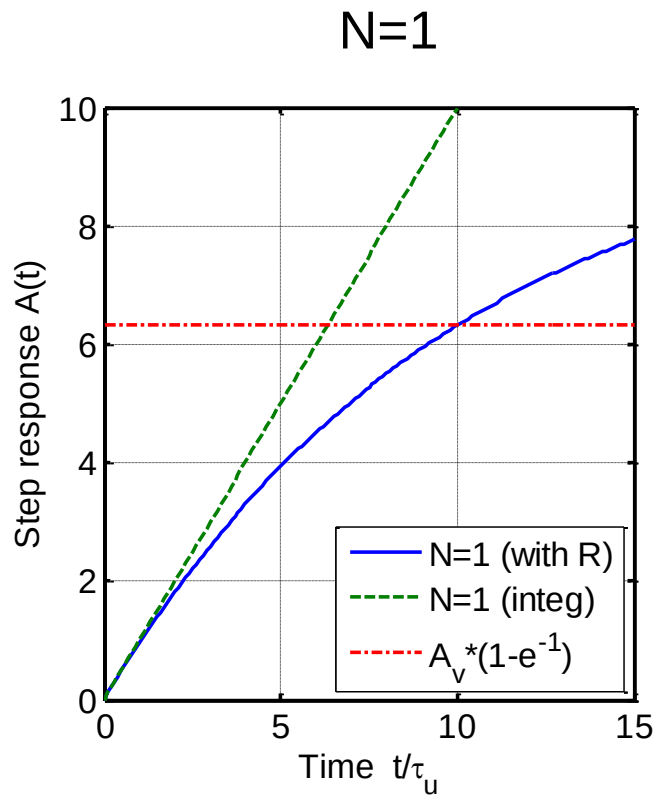
$$v_{o1} = \frac{g_m}{sC} v_{in} = \frac{\omega_u}{s} v_{in} \Rightarrow v_{oN} = \frac{\omega_u^N}{s^N} v_{in}$$

$$V_{out}(s) = \frac{V_{step}}{s} \frac{\omega_u^N}{s^N}$$

$$V_{out}(t) = V_{step} \omega_u^N \frac{t^N}{N!}$$

$$A(\tau_d) = \omega_u^N \frac{t^N}{N!}$$

Simulation Results



Analysis

- Cascade of integrators is faster than cascade of resistively loaded stages
- Integrator requires reset, but not a problem due to clocked comparison
- Delay time

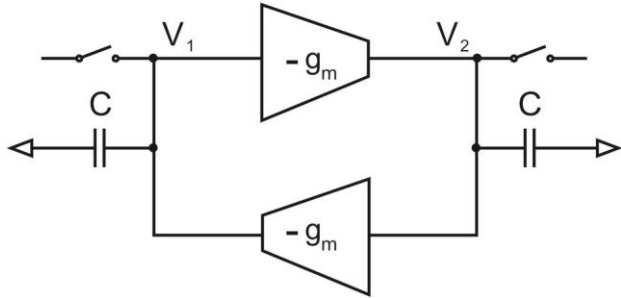
$$\tau_d = \tau_u [N! A(\tau_d)]^{1/N}$$

- Optimum number of stages approximately given by

$$N_{\text{opt}} = 1.1 \ln[A(\tau_d)] + 0.79 \quad [\text{Wu, JSSC 12/1988}]$$

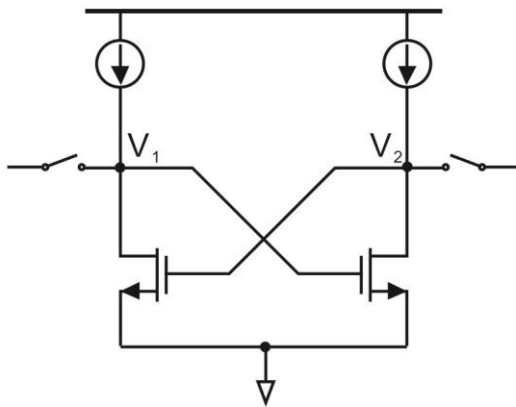
- Effective gain per stage is still close to $e \approx 2.7183$
- Can we further increase the speed but without requiring many stages?

Use Positive Feedback – Regenerative Latch



$t < 0$ setup initial condition $v_{10} - v_{20} = v_{d0}$

$t \geq 0$ enable positive feedback

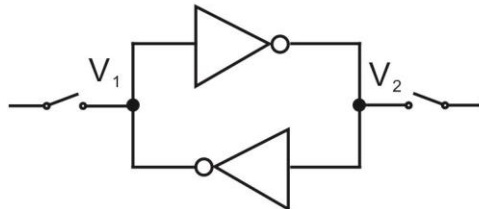


$$\begin{cases} \frac{dv_1}{dt} = \frac{i_1(t)}{C} = \frac{-g_m v_2(t)}{C} \\ \frac{dv_2}{dt} = \frac{i_2(t)}{C} = \frac{-g_m v_1(t)}{C} \end{cases}$$

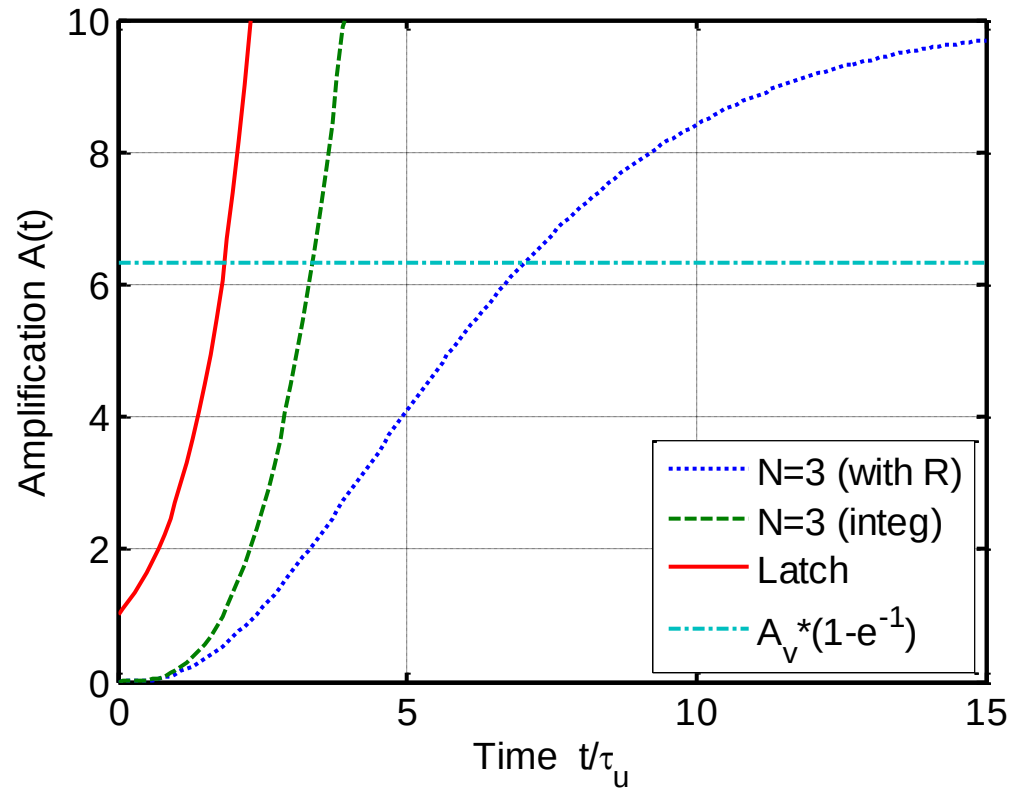
$$\Rightarrow v_1(t) - v_2(t) = v_d(t) = v_{d0} e^{t/\tau_u}$$

$$\Rightarrow A(t) = \frac{v_d(t)}{v_{d0}} = e^{t/\tau_u}$$

$$\tau_u = \frac{C}{g_m}$$



Comparison



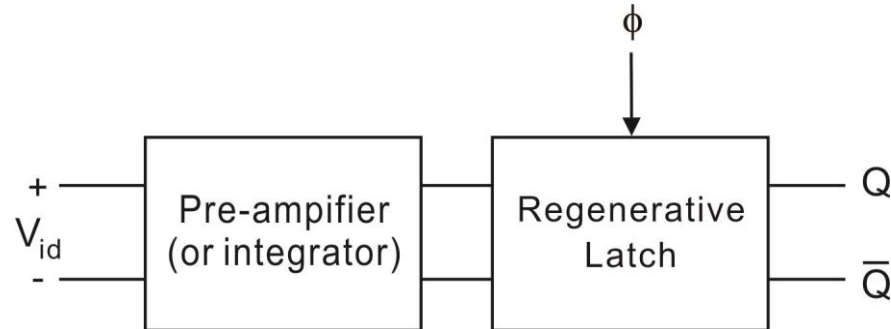
- Exponential wins over polynomial!
- Latch is much faster than cascade of amplifiers/integrators

Latch "Gain"

$A(\tau_d)$	τ_d/τ_u
10	2.3
100	4.6
1,000	6.9
10,000	9.2

- Trade exponential gain (A) with linear delay (τ_d)
- Better than amplifier and integrator

Standard Comparator Architecture



- Why bother using pre-amplification?
 - Offset
 - Use pre-amplification to lower input referred offset
 - Hard to build latches with offset $< 10\sim 100\text{mV}$
 - Common mode rejection
 - Comparator offset can be input dependent
 - Attenuate "kickback noise"
 - Metastability

Metastability (1)

- Consider minimum initial latch input voltage needed to regenerate to V_{DD} within maximum available time $T_{\max}$

$$V_{d0\min} = \frac{V_{DD}}{e^{T_{\max}/\tau_u}}$$

- Minimum-required pre-amplifier input

$$V_{id0\min} = \frac{1}{A_V} \frac{V_{DD}}{e^{T_{\max}/\tau_u}}$$

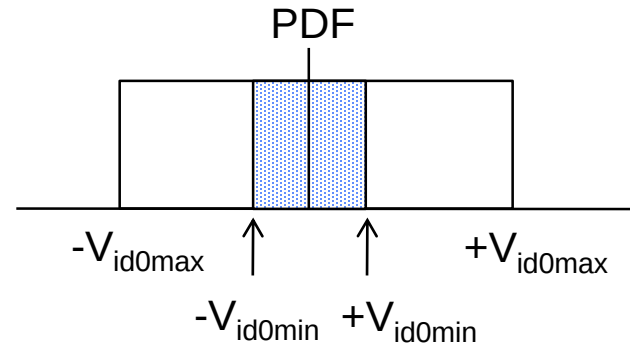
- Probability of seeing a metastable output

$$P(\text{Error}) = P(V_{id0} < V_{id0\min})$$

Metastability (2)

- Assuming a uniform input signal distribution over some range, we have

$$P(\text{Error}) = \frac{V_{id0\min}}{V_{id0\max}}$$



- In a flash ADC, only 1 comparator can be metastable. The effective input range $V_{id0\max}$ over which the signal is distributed is thus $\Delta/2$

$$P(\text{Error}) = \frac{\frac{1}{A_V} V_{DD} e^{-T_{\max} / \tau_u}}{\frac{\Delta}{2}} = \frac{2^{B+1}}{A_V} \frac{V_{DD}}{V_{FS}} e^{-T_{\max} / \tau_u}$$

Metastability (3)

- Example: 6-bit, 1GHz Flash ADC, $T_{\max}=T_s/2=0.5\text{ns}$, $\tau_u=1/(2\pi\cdot 10\text{GHz})=16\text{ps}$, $A_v=3$, $V_{\text{FS}}=V_{\text{DD}}$

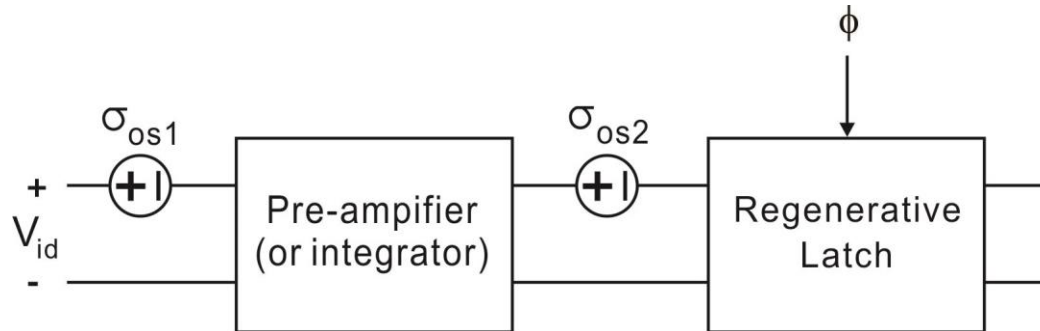
$$P(\text{Error}) = \frac{2^{6+1}}{3} e^{-500/16} \approx 10^{-12}$$

- Mean time to failure (MTF)

$$MTF = \frac{1}{P(\text{Error}) f_s} = \frac{1}{10^{-12} \times 1 \times 10^9} \text{sec} = 1000 \text{sec} \approx 17 \text{min}$$

- We can enlarge MTF by
 - Reducing speed (larger $T_{\max}/\tau_u$)
 - Exponential dependence
 - Increasing pre-amplifier gain A_v
 - Linear dependence

Comparator Offset



$$\sigma_{OS}^2 = \sigma_{OS1}^2 + \frac{1}{A_V^2} \sigma_{OS2}^2$$

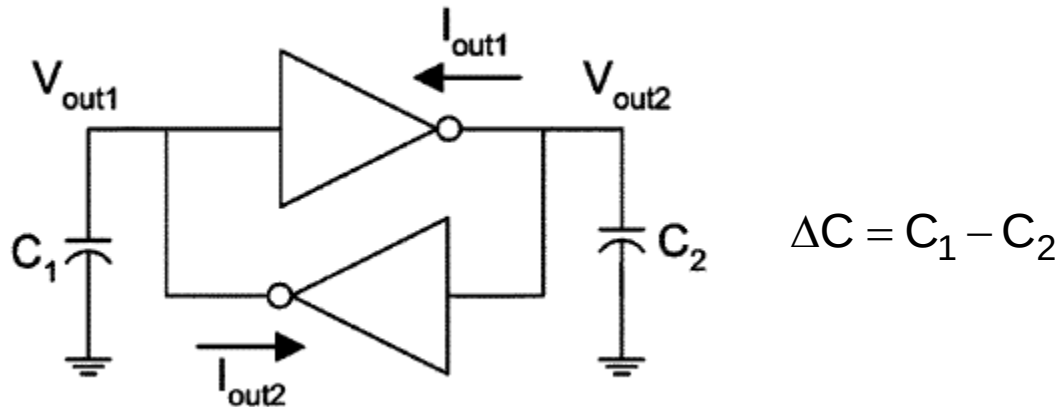
- Example: $\sigma_{VOS1}=1\text{mV}$, $\sigma_{VOS2}=10\text{mV}$, $A_V=10$

$$\sigma_{VOS} = \sqrt{(1\text{mV})^2 + \frac{1}{10^2} (10\text{mV})^2} = 1.4\text{mV}$$

Causes for Comparator Offset

- V_{th} mismatch
- Device dimension mismatch
- Load capacitance mismatch
 - Load both sides even if only one side output is needed
 - Be careful about wiring

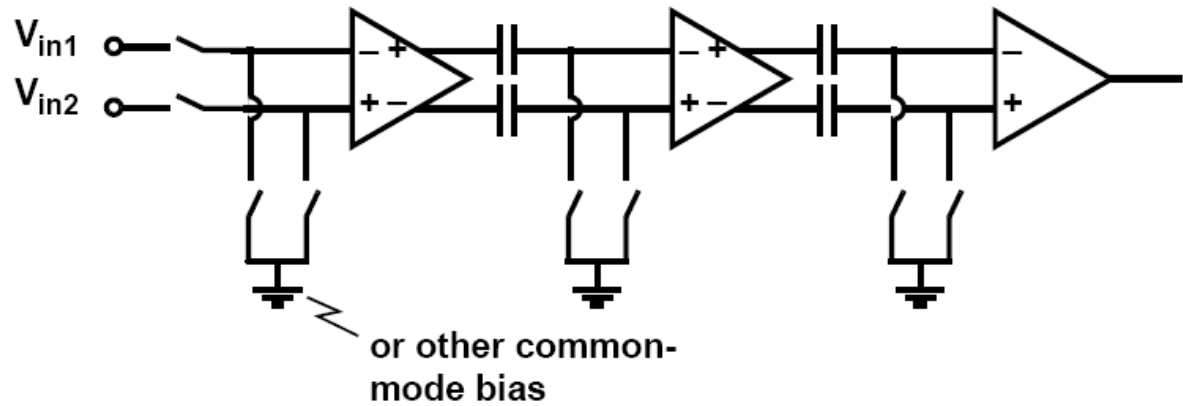
Load Capacitance Mismatch



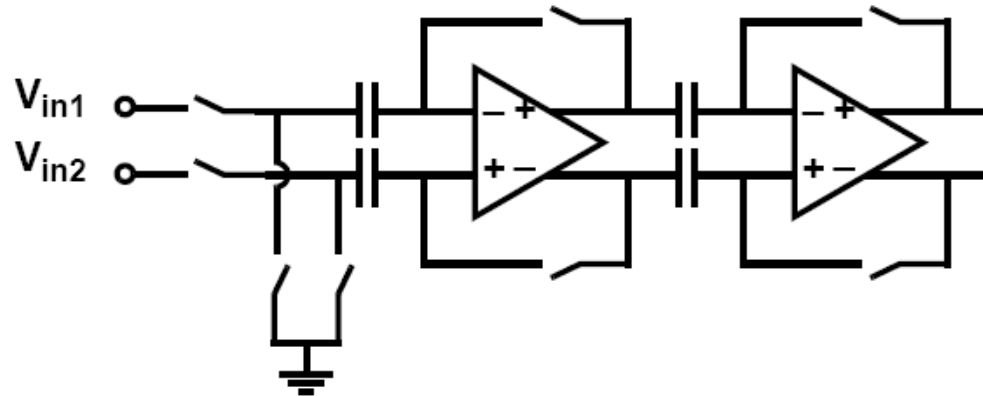
- Can show $V_{os} \cong 0.5(V_{(t=0)} - V_S)\Delta C/C$; V_S = inverter switching voltage
 - Nikoozadeh & Murmann, IEEE TCAS II, Dec. 2006.
- Example
 - $0.5 \cdot 10\text{fF}/100\text{fF} \cdot (1\text{V} - 0.5\text{V}) = 25\text{mV}$ (!)
- Need to match the output load.

Preamplifier Offset Cancellation

Output Series Cancellation

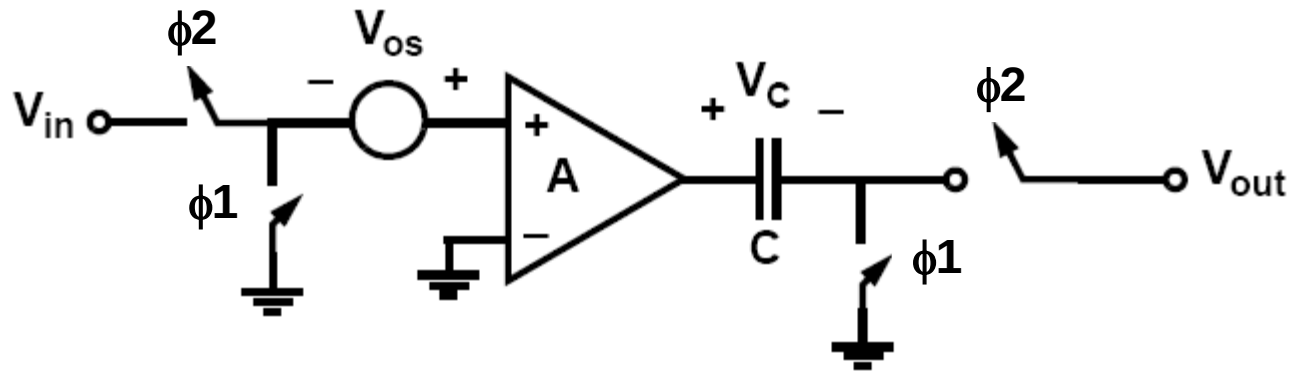


Input Series Cancellation



Output Series Cancellation

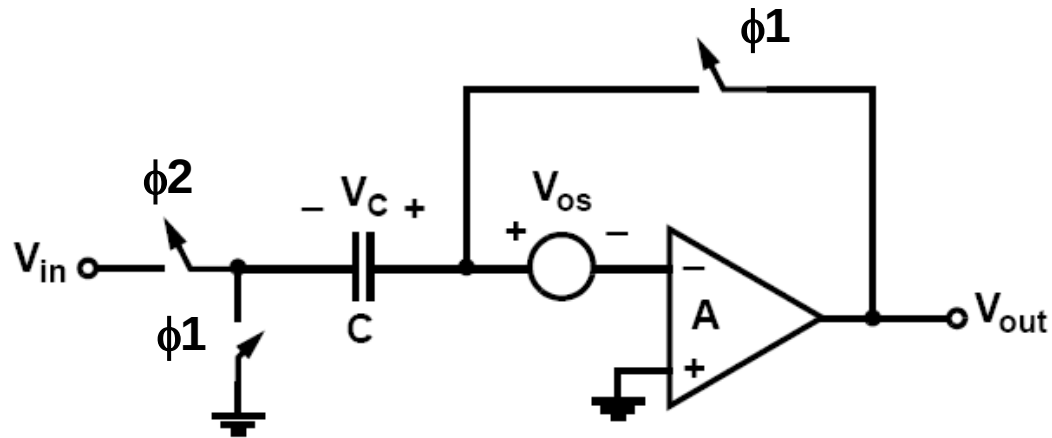
Ref: Poujois, et al., JSSC 8/78



- Phase 1: Offset storage; Phase 2: Amplify
- Design considerations
 - Must ensure that amplifier does not saturate during phase 1
 - Must make C sufficiently large to avoid attenuation and mitigate charge injection error

Input Series Cancellation

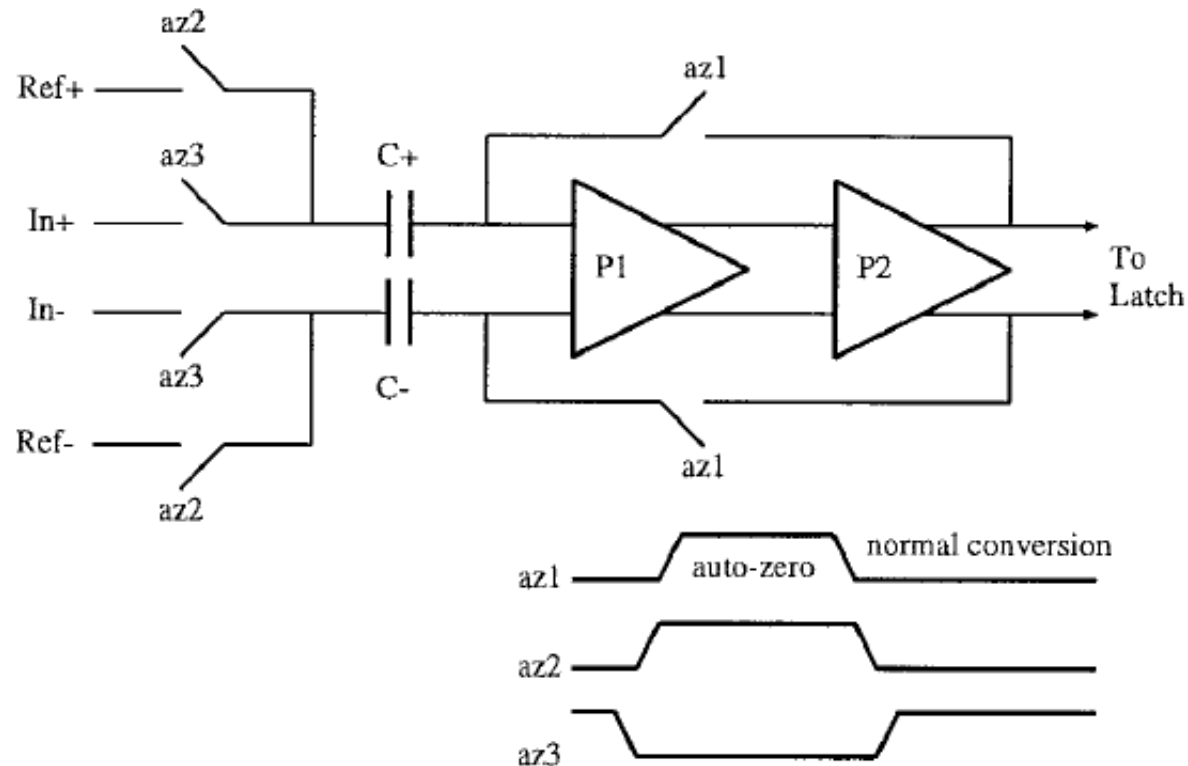
Refs: McCreary & Gray
Yee, et al.



- Phase 1: Offset storage; Phase 2: Amplify
- In phase 2, input referred offset $\approx V_{os}/(A+1)$
 - 4x reduction if $A=3$

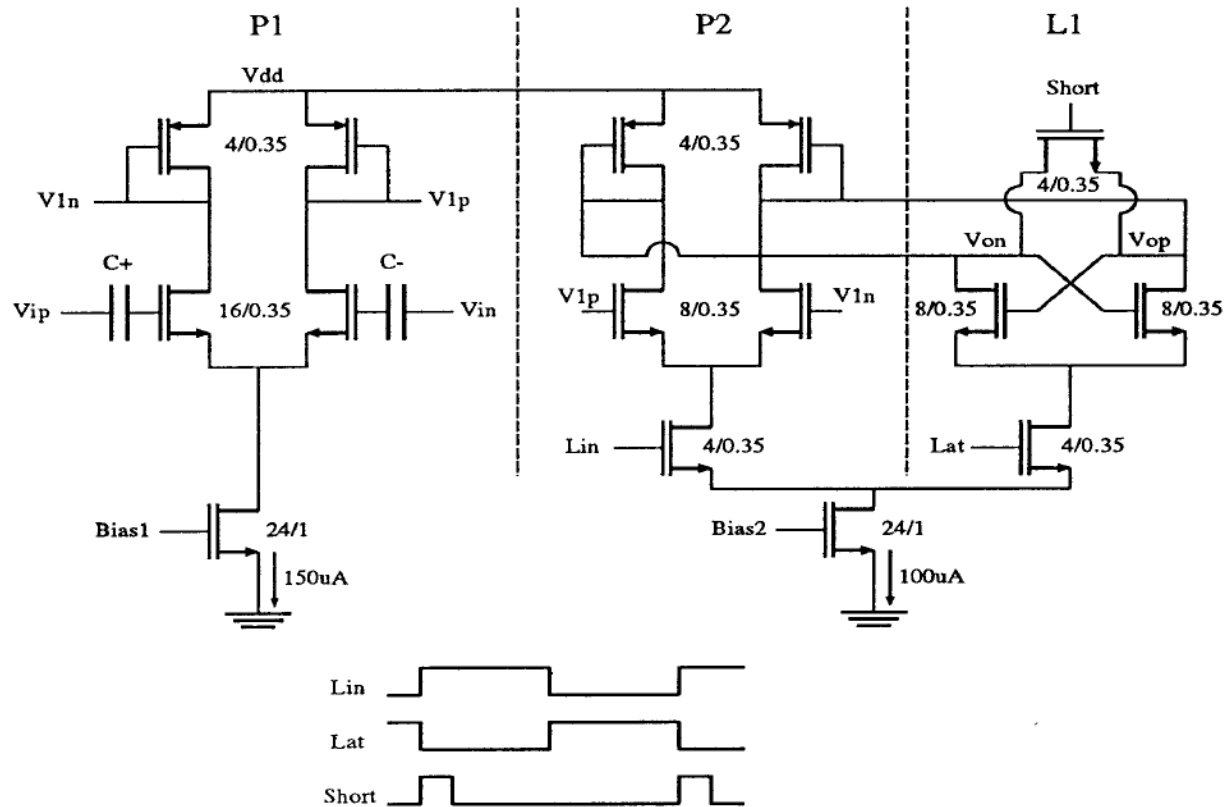
Comparator Example #1

- Mehr & Dalton, JSSC 7/1999



Comparator Example #1

- Mehr & Dalton, JSSC 7/1999



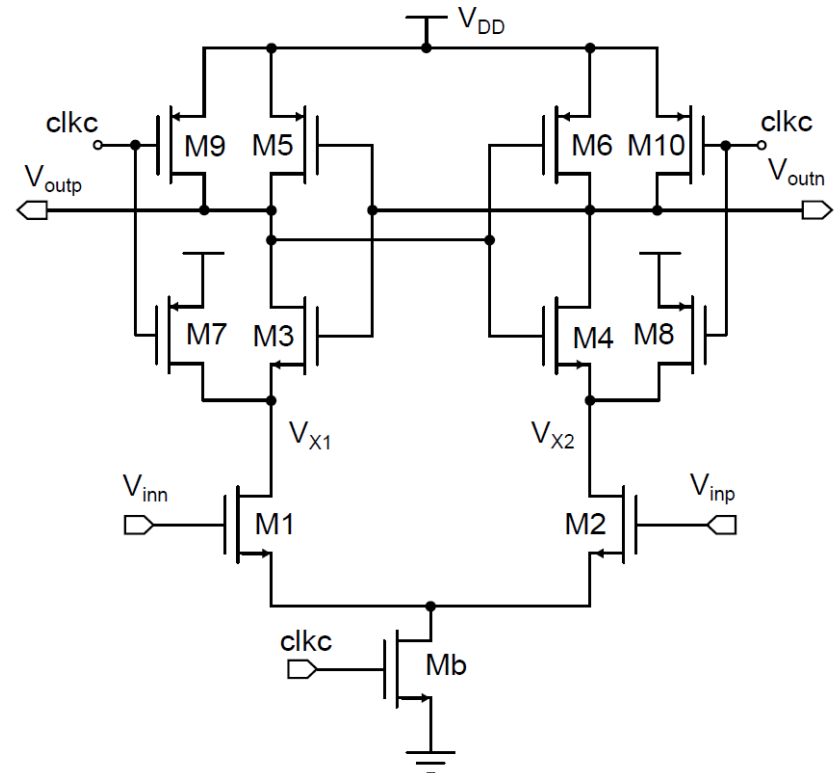
- Two stage gain; current re-use; short the output to clear charge

- Input series cancellation; create input common mode; couple the current



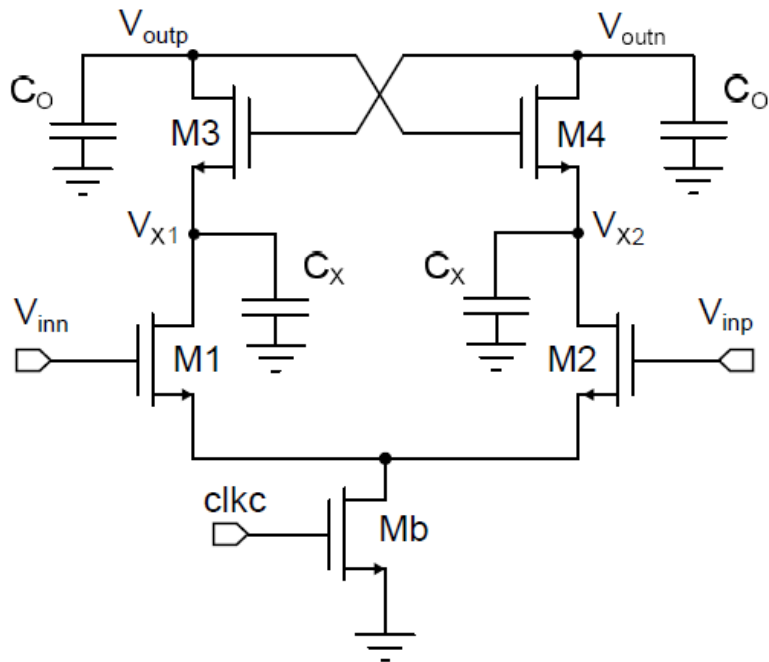
Comparator Example #3

- Dynamic comparator
- Also called StrongArm latch
- Low power, no DC current!
- Have inherent dynamic preamplifier

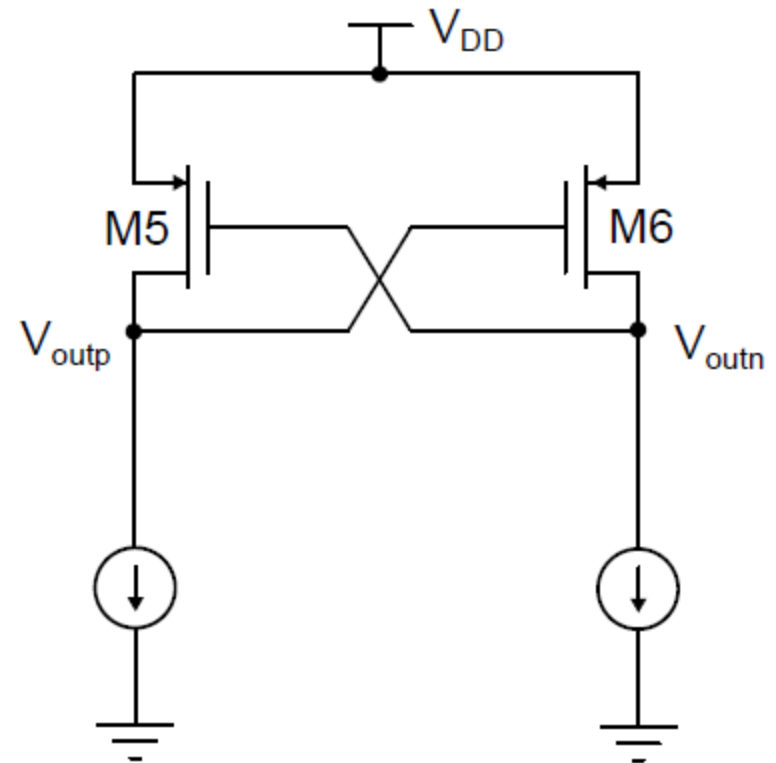


Latch Operation

Pre-amplification phase

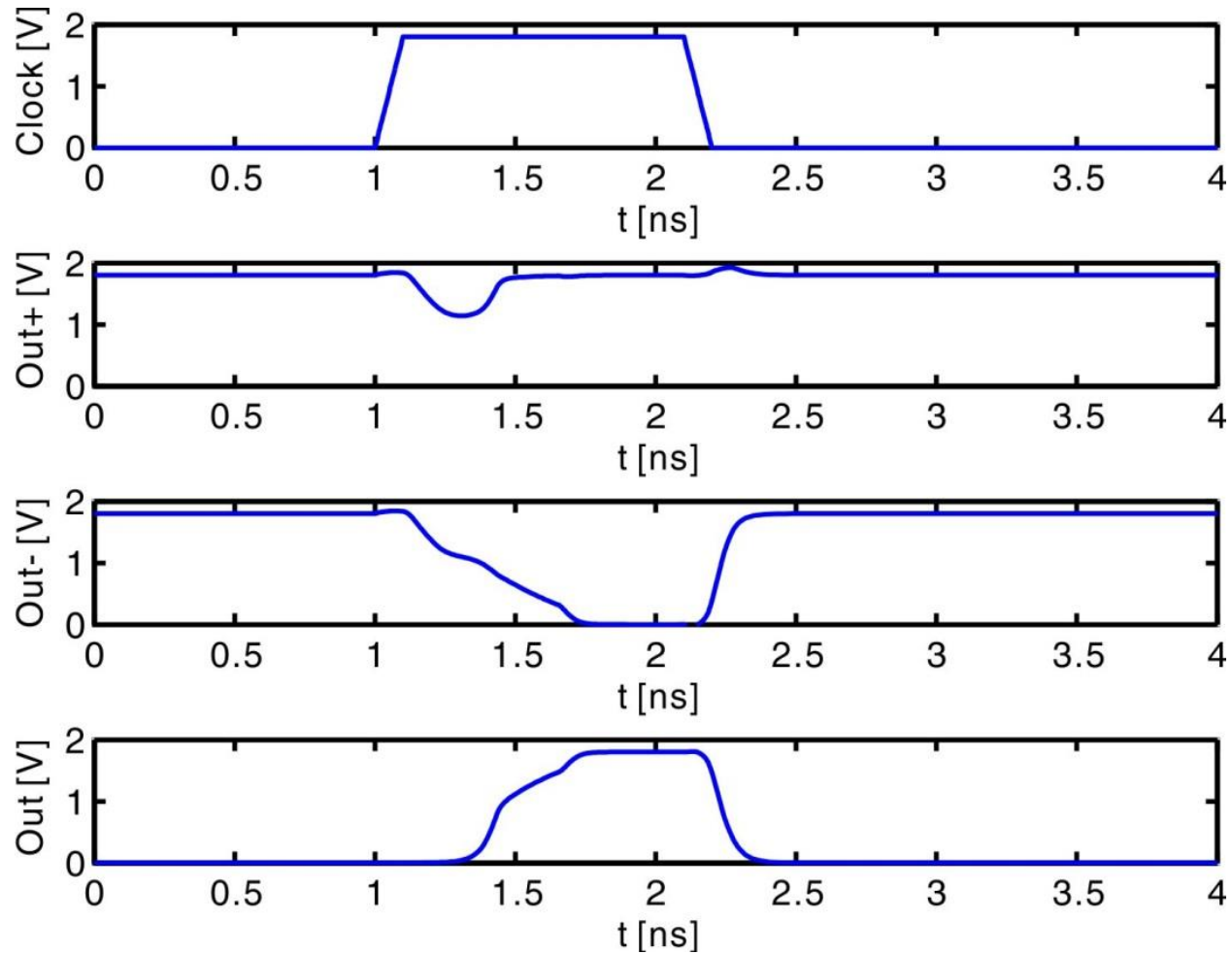


Self regeneration phase

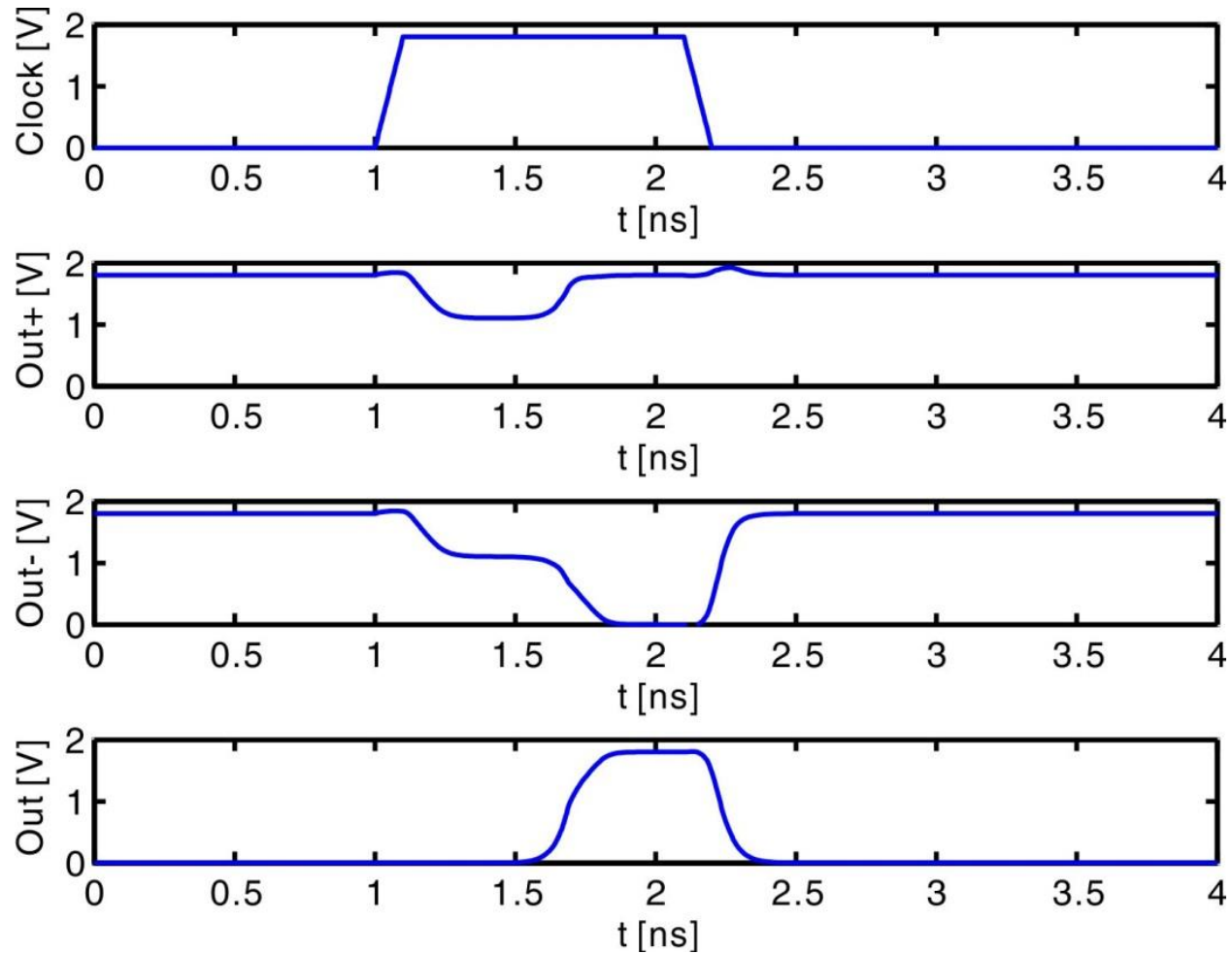


$$G \cong \frac{\Delta V_{out}}{\Delta V} \cong \frac{\Delta Q / C_O}{\Delta V} \cong \frac{g_m}{I_D} \left\{ V_{Tp5,6} + \frac{C_X}{C_O} (V_{Tp5,6} + V_{Tn3,4}) \right\}$$

Simulation for $V_{in} = 1\text{mV}$

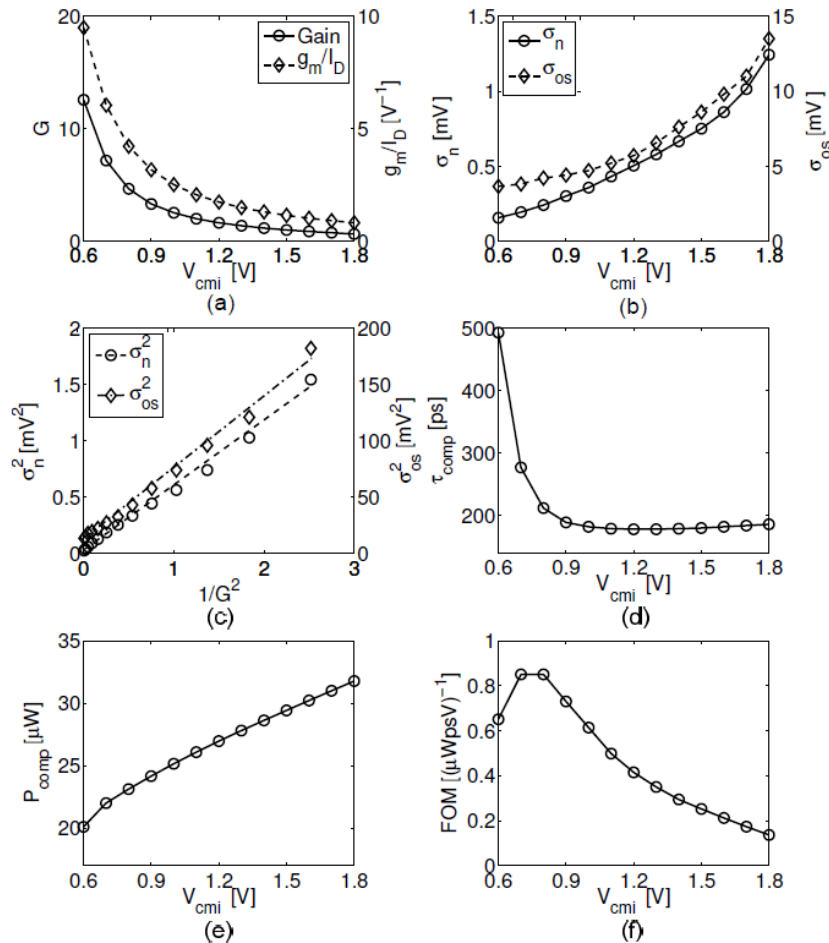


Simulation for $V_{in} = 1\mu V$



Design tradeoff

- Preamplifier gain G is proportional to g_m/I_D .
 - For large G , we prefer a small V_{cm} .
- Comparator offset and noise increases as G decreases.
- For high speed, we prefer a large V_{cm} and smaller g_m/I_D .
- Comparator power increases gradually as V_{cm} increases.



Two-Stage Implementation (1)

- Claimed to work better at low voltage and have a faster speed
 - Be careful about timing mismatches between Clk and Clkb

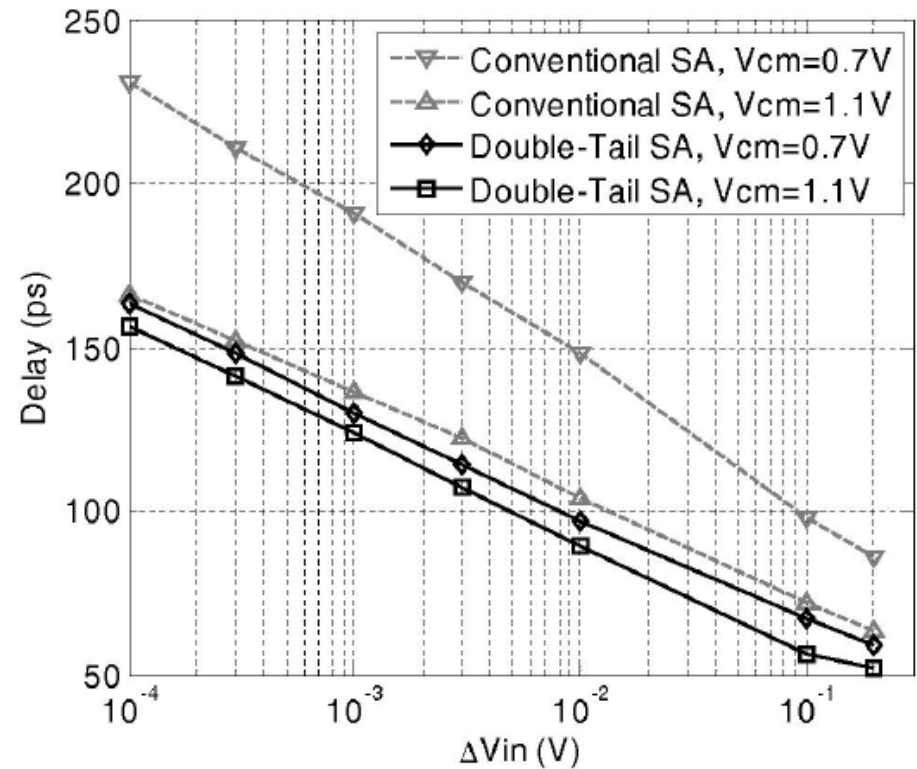
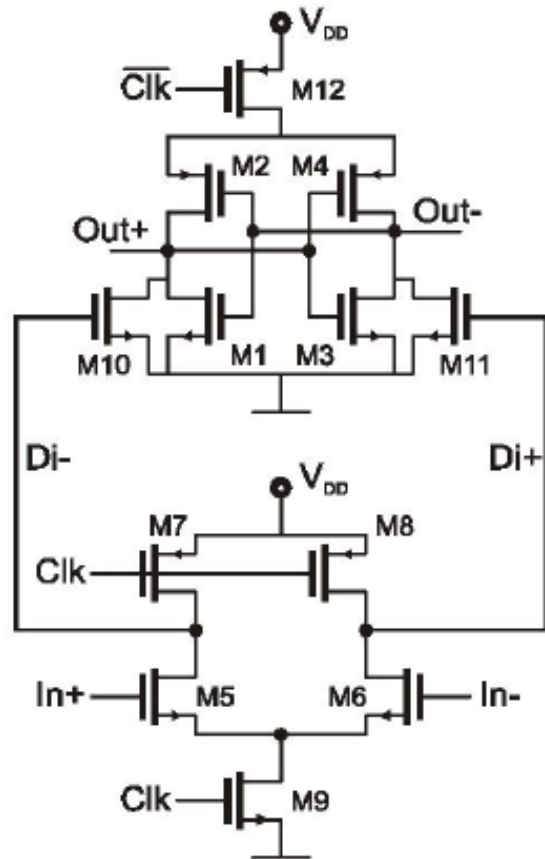
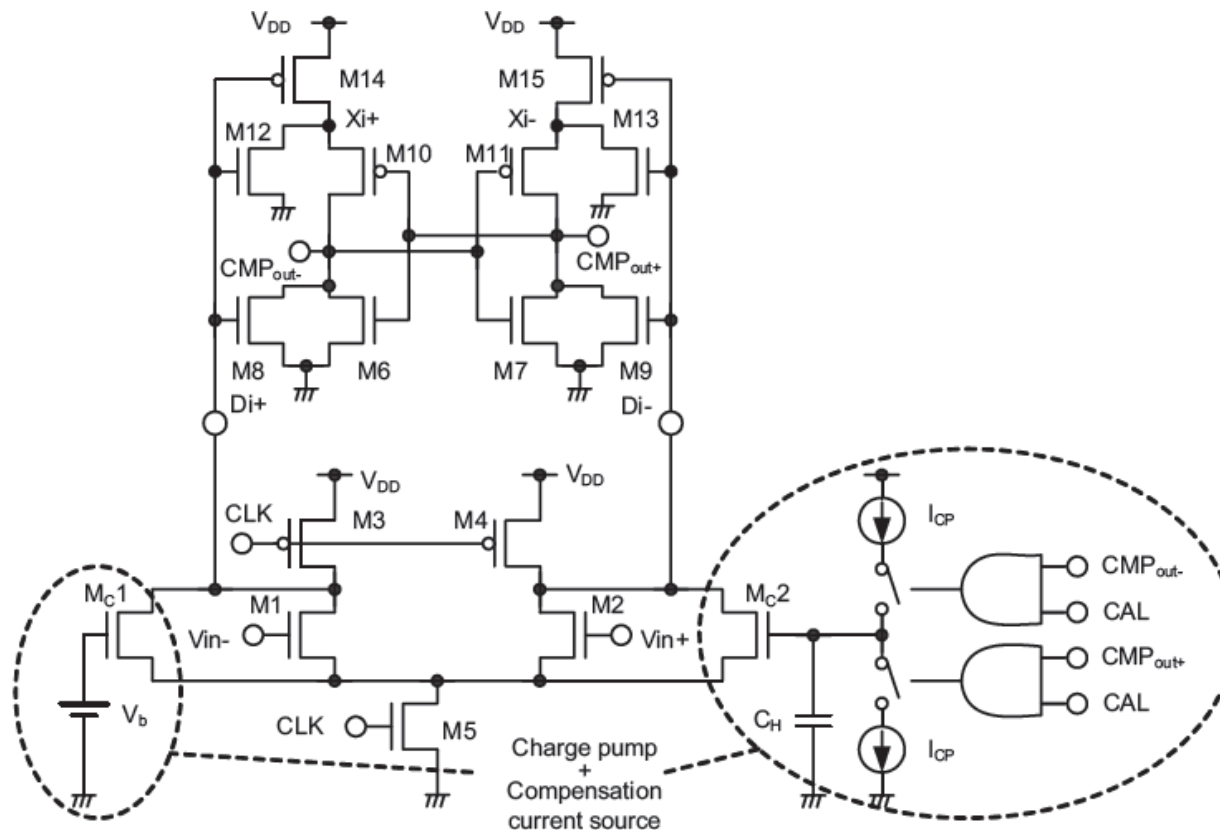


Figure 17.7.3: Simulated sense amplifier delays versus differential input voltage. The delay is the time between the clock edge and the instant when ΔOut crosses $1/2 V_{DD}$.

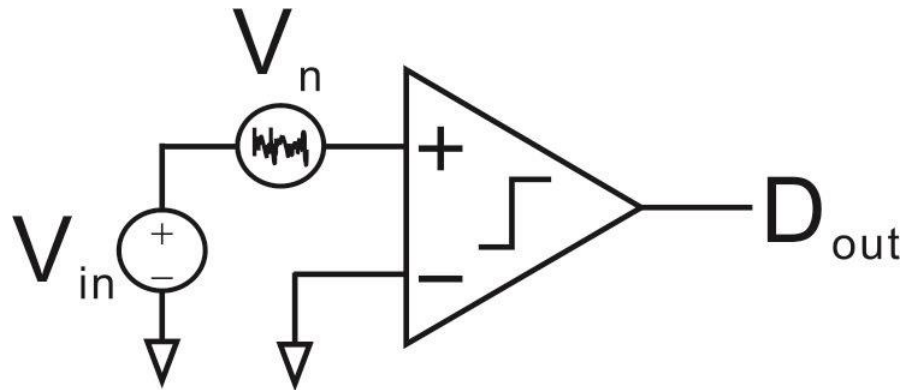
Two-Stage Implementation (2)

- Require only 1 clock
- M14 and M15 also serve as input transistors, which provide extra gain.

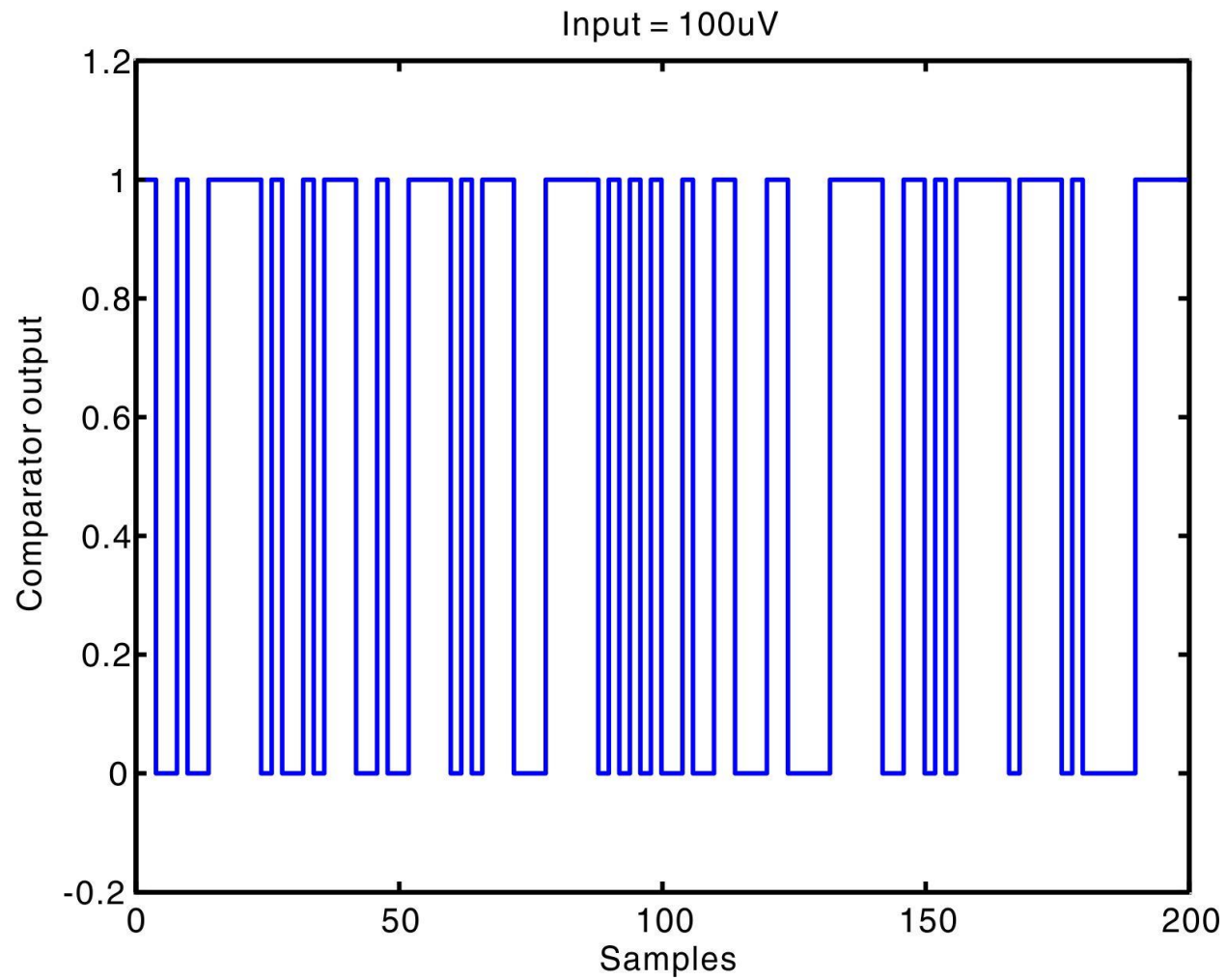


Comparator Noise Simulation (1)

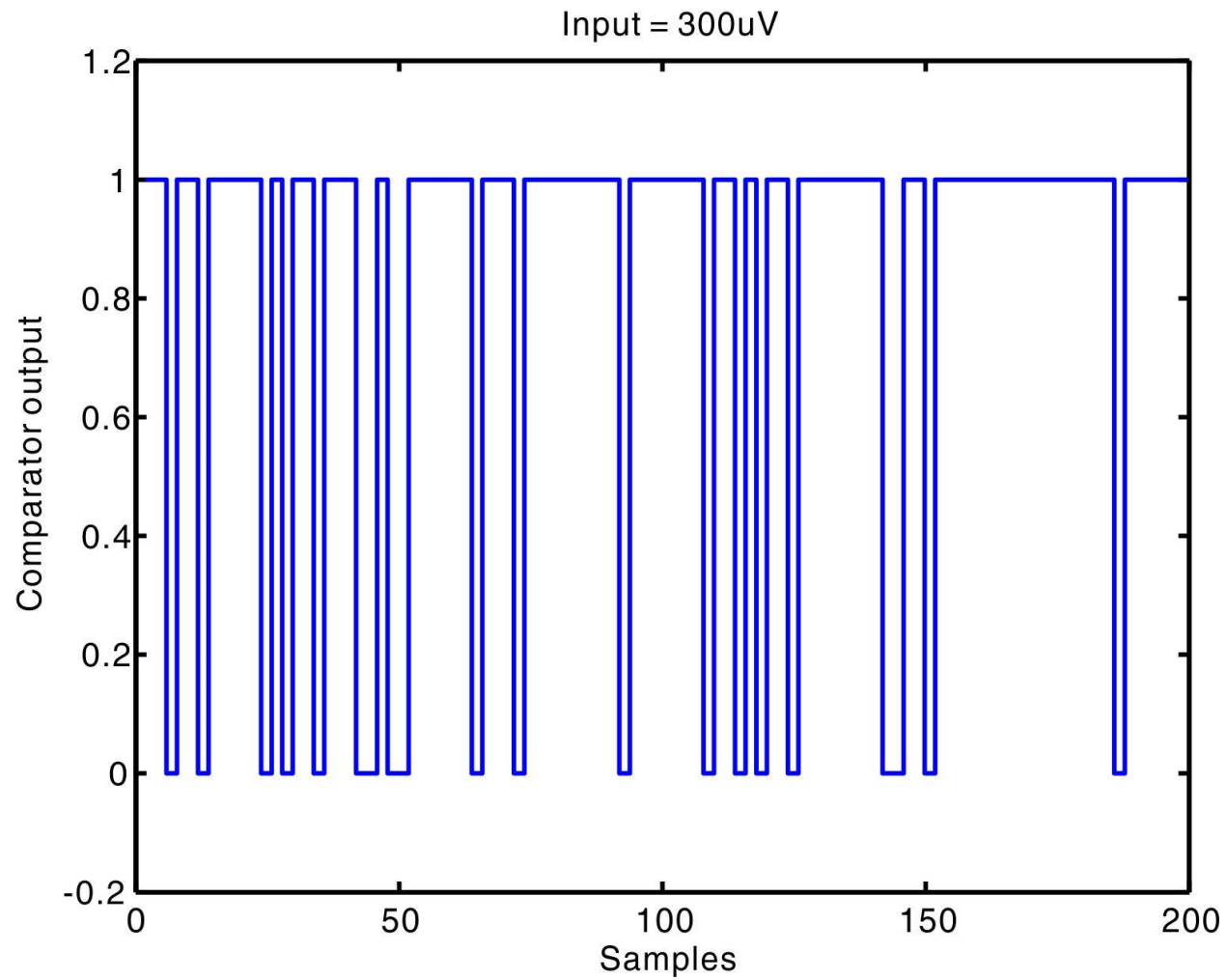
- Comparator is a dynamic circuit
 - AC noise simulation does not apply, as there is no operating point
 - PSS noise simulation does not apply either, as there is no steady-state
- Transient noise simulation is best option
 - We give an nonzero input, and calculate the input referred comparator noise based on the histogram



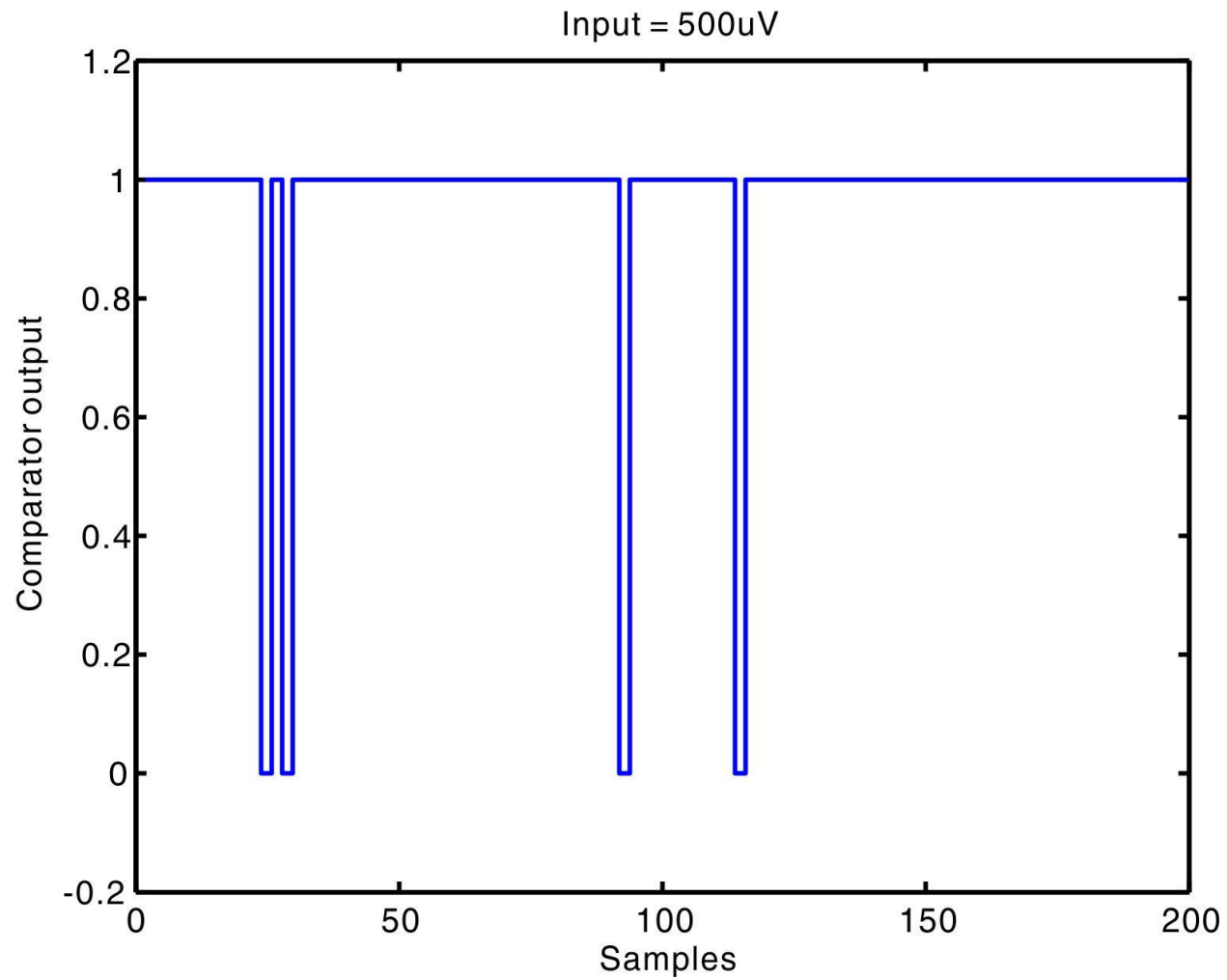
$$V_{in} = 100 \text{ } \mu\text{V}$$



$$V_{in} = 300 \text{ } \mu\text{V}$$



$$V_{in} = 500 \mu V$$



Comparator Noise Simulation (2)

- Assuming comparator input referred noise follows normal distribution, we can estimate comparator noise σ based on the following formula:

$$P(D_{out} = 1) = P(V_n < V_{in}) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{V_{in}} e^{-\frac{x^2}{2\sigma^2}} dx$$

- $V_{in} = 300\mu V$, $P(D_{out}=1) = 80\% \rightarrow \sigma \approx 350\mu V$
 - In MATLAB, $\sigma = V_{in} / \text{norminv}(P, 0, 1)$
- To reduce estimation error, we want to:
 - Simulate more samples
 - Make $V_{in} \approx \sigma$
- Be careful that here we assume that the comparator has no offset.
 - If the comparator has systematic offset, we need to take it away first